

■ AI Data Analyst - Executive Analysis Report

■ Executive Data Analysis Report

1. Executive Summary

- The **Butchers** category generates the highest revenue (\$215,930) while **Milk Products** has the lowest average spend per transaction.
- Monthly spend is relatively stable across the 3■year window, with a peak in **January 2022** (\$55,298) and a recent low in **January 2025** (\$26,742).
- Payment methods are evenly distributed (≈4,100 – 4,300 transactions each), indicating no single channel dominance.
- Discounts are applied in ~65■70 % of transactions, highest in **Computers & electric accessories** (69.38 %).
- The “**Unknown**” item accounts for **\$144,606.5** of total spend, suggesting a data■quality issue that should be investigated.

These findings answer the request for a **Power BI■style executive dashboard** by providing a primary■category breakdown, trend analysis, payment■method distribution, discount utilization, and top■selling items.

2. Dataset Overview & Data Quality Profile

Aspect Detail
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Rows × Columns 12,575 × 11
Duplicate rows 0
Missing values Item, Price Per Unit, Quantity, Total Spent, Discount Applied
Potential date column Transaction Date (originally string)
Data quality notes Missing values were present in key numeric and categorical fields; date needed conversion.

3. Data Cleaning & Validation Summary

Step Action Result

|-----|-----|-----|

| 1 | Copied original **df** to **result** | Original data untouched |

| 2 | Dropped duplicates (none) | No change |

| 3 | Stripped whitespace from all object columns | Clean strings |

| 4 | Converted **Transaction Date** → **datetime** (coerce errors) | Proper date type |

| 5a | Filled missing **Item** with "Unknown" | No nulls in **Item** |

| 5b | Filled numeric columns (**Price Per Unit**, **Quantity**, **Total Spent**) with median values | Numeric completeness |

| 5c | Filled **Discount Applied** with **False** and cast to **bool** | Boolean consistency |

| 6 | Re■calculated any missing **Total Spent** as **Price Per Unit × Quantity** | Financial integrity |

| 7 | Cast **Total Spent** to **float** | Uniform numeric type |

| **Validation** | All rows (12,575) now clean and ready for analysis | ■ |

4. User Query & Analytical Approach

User question:

> “Generate a complete PowerBI■style executive dashboard overview with primary category breakdown and trends.”

Analytical methodology (implemented in Pandas):

1. **Category breakdown** – grouped by **Category** to compute total spend, distinct transaction count, average spend, and average quantity.
2. **Monthly trend** – extracted calendar month from **Transaction Date**, then summed spend and counted distinct transactions per month.
3. **Payment■method distribution** – counted occurrences of each **Payment Method**.
4. **Discount usage rate** – calculated the mean of **Discount Applied** per category (interpreted as % of transactions with a discount).
5. **Top■5 revenue items** – summed **Total Spent** per **Item** and selected the five highest.

All results were converted to JSON■serialisable dictionaries for easy downstream consumption.

5. Python Analysis Code & Executed Data Output

Code

```
# Convert the transaction date column to datetime (working on a copy) df_dates = df.copy()
df_dates['Transaction Date'] = pd.to_datetime(df_dates['Transaction Date']) # 1. Category
breakdown category_summary_df = ( df_dates.groupby('Category') .agg( total_spent=('Total
Spent', 'sum'), transaction_count=('Transaction ID', 'nunique'), avg_spent=('Total Spent',
'mean'), avg_quantity=('Quantity', 'mean') ) .reset_index() ) # 2. Monthly trends (by calendar
month) df_dates['Month'] = df_dates['Transaction Date'].dt.to_period('M').astype(str)
monthly_trend_df = ( df_dates.groupby('Month') .agg( total_spent=('Total Spent', 'sum'),
transactions=('Transaction ID', 'nunique') ) .reset_index() .sort_values('Month') ) # 3.
Payment method distribution (overall) payment_method_distribution_df = ( df_dates['Payment
Method'] .value_counts() .reset_index() .rename(columns={'index': 'payment_method', 'Payment
Method': 'count'}) ) # 4. Discount usage rate per category discount_rate_by_category_df = (
df_dates.groupby('Category')['Discount Applied'] .mean() .reset_index()
.rename(columns={'Discount Applied': 'discount_rate'}) ) # 5. Top 5 items by revenue
top_items_df = ( df_dates.groupby('Item')['Total Spent'] .sum() .nlargest(5) .reset_index()
.rename(columns={'Total Spent': 'total_spent'}) ) # Assemble JSON■serialisable result result =
{ 'category_summary': category_summary_df.to_dict(orient='records'), 'monthly_trend':
monthly_trend_df.to_dict(orient='records'), 'payment_method_distribution':
payment_method_distribution_df.to_dict(orient='records'), 'discount_rate_by_category':
discount_rate_by_category_df.to_dict(orient='records'), 'top_items':
top_items_df.to_dict(orient='records') }
```

Executed Data Output (JSON)

```
{ "category_summary": [ {"Category": "Beverages", "total_spent": 204751.0, "transaction_count": 156
7, "avg_spent": 130.6643267389917, "avg_quantity": 5.6056158264199105}, {"Category": "Butchers", "to
tal_spent": 215930.0, "transaction_count": 1568, "avg_spent": 137.71045918367346, "avg_quantity": 5.5
08928571428571}, {"Category": "Computers and electric accessories", "total_spent": 199481.0, "tran
saction_count": 1558, "avg_spent": 128.03658536585365, "avg_quantity": 5.621309370988446},
{"Category": "Electric household essentials", "total_spent": 211951.0, "transaction_count": 1591, "a
vg_spent": 133.21873035826525, "avg_quantity": 5.50534255185418}, {"Category": "Food", "total_spent
": 203600.5, "transaction_count": 1588, "avg_spent": 128.21190176322418, "avg_quantity": 5.5875314861
46096}, {"Category": "Furniture", "total_spent": 202471.0, "transaction_count": 1591, "avg_spent": 12
7.26021370207417, "avg_quantity": 5.5675675675675675}, {"Category": "Milk Products", "total_spent"
: 187815.5, "transaction_count": 1584, "avg_spent": 118.57039141414141, "avg_quantity": 5.53345959595
9596}, {"Category": "Patisserie", "total_spent": 191605.0, "transaction_count": 1528, "avg_spent": 12
5.39594240837697, "avg_quantity": 5.539921465968586} ], "monthly_trend": [
{"Month": "2022-01", "total_spent": 55298.5, "transactions": 390},
{"Month": "2022-02", "total_spent": 45278.5, "transactions": 331},
{"Month": "2022-03", "total_spent": 42515.0, "transactions": 353},
{"Month": "2022-04", "total_spent": 41744.0, "transactions": 346},
{"Month": "2022-05", "total_spent": 41975.0, "transactions": 330},
{"Month": "2022-06", "total_spent": 44420.5, "transactions": 338},
{"Month": "2022-07", "total_spent": 46858.5, "transactions": 371},
{"Month": "2022-08", "total_spent": 43069.5, "transactions": 347},
{"Month": "2022-09", "total_spent": 47307.0, "transactions": 348},
{"Month": "2022-10", "total_spent": 39874.0, "transactions": 317},
{"Month": "2022-11", "total_spent": 44077.5, "transactions": 354},
{"Month": "2022-12", "total_spent": 40588.0, "transactions": 309},
{"Month": "2023-01", "total_spent": 49246.0, "transactions": 373},
{"Month": "2023-02", "total_spent": 40842.0, "transactions": 314},
{"Month": "2023-03", "total_spent": 40813.0, "transactions": 311},
{"Month": "2023-04", "total_spent": 40533.0, "transactions": 323},
{"Month": "2023-05", "total_spent": 42216.5, "transactions": 334},
{"Month": "2023-06", "total_spent": 43884.5, "transactions": 349},
{"Month": "2023-07", "total_spent": 47694.0, "transactions": 375},
{"Month": "2023-08", "total_spent": 39677.0, "transactions": 317},
{"Month": "2023-09", "total_spent": 42588.0, "transactions": 333},
{"Month": "2023-10", "total_spent": 39949.5, "transactions": 308},
{"Month": "2023-11", "total_spent": 38316.0, "transactions": 302},
{"Month": "2023-12", "total_spent": 44974.0, "transactions": 348},
{"Month": "2024-01", "total_spent": 50295.5, "transactions": 385},
```

```
{
  "Month": "2024-02", "total_spent": 38881.0, "transactions": 320,
  "Month": "2024-03", "total_spent": 44923.0, "transactions": 355,
  "Month": "2024-04", "total_spent": 48007.0, "transactions": 327,
  "Month": "2024-05", "total_spent": 46370.5, "transactions": 369,
  "Month": "2024-06", "total_spent": 46565.5, "transactions": 351,
  "Month": "2024-07", "total_spent": 42381.5, "transactions": 343,
  "Month": "2024-08", "total_spent": 45640.5, "transactions": 375,
  "Month": "2024-09", "total_spent": 43463.5, "transactions": 340,
  "Month": "2024-10", "total_spent": 44364.0, "transactions": 353,
  "Month": "2024-11", "total_spent": 45703.5, "transactions": 347,
  "Month": "2024-12", "total_spent": 50528.0, "transactions": 376,
  "Month": "2025-01", "total_spent": 26742.0, "transactions": 213 } ],
  "payment_method_distribution": [
    { "count": 4310 }, { "count": 4144 }, { "count": 4121 } ],
  "discount_rate_by_category": [
    { "Category": "Beverages", "discount_rate": 0.6483726866624122 },
    { "Category": "Butchers", "discount_rate": 0.6741071428571429 },
    { "Category": "Computers and electric accessories", "discount_rate": 0.6938382541720154 },
    { "Category": "Electric household essentials", "discount_rate": 0.6599622878692646 },
    { "Category": "Food", "discount_rate": 0.6643576826196473 },
    { "Category": "Furniture", "discount_rate": 0.6417347580138277 },
    { "Category": "Milk Products", "discount_rate": 0.6875 },
    { "Category": "Patisserie", "discount_rate": 0.6590314136125655 } ],
  "top_items": [
    { "Item": "Unknown", "total_spent": 144606.5 },
    { "Item": "Item_25_FUR", "total_spent": 25256.0 },
    { "Item": "Item_25_EHE", "total_spent": 23083.0 },
    { "Item": "Item_25_BUT", "total_spent": 21894.0 },
    { "Item": "Item_24_FUR", "total_spent": 21172.0 } ] }

```

6. Key Insights & Strategic Recommendations

> ■ Insight 1 – Revenue Concentration

> *Butchers* leads category revenue (\$215.9 K). Focus marketing spend and inventory planning on this high-margin segment to sustain growth.

> ■ Insight 2 – Discount Overuse

> Discounts are applied in ~65-70 % of transactions, especially in *Computers & electric accessories* (69.38 %). Consider tightening discount policies or introducing tiered promotions to protect margins.

> ■ Insight 3 – Stable Monthly Performance

> Monthly spend hovers around \$40-\$50 K with modest seasonal variation. The dip in **Jan 2025** may signal a post-holiday slowdown; plan inventory reductions or targeted campaigns for early-year periods.

> ■ Insight 4 – Payment Method Parity

> No single payment method dominates. This suggests the current mix meets customer preferences, but there may be an opportunity to negotiate better interchange rates with